

Practice Assignment on Testing, Linux and Servers

Task 1: System Monitoring Setup

Step 1: Install Monitoring Tool (htop)

```
vinamra@vinamra-mac ~ % brew --version
Homebrew 4.5.13
vinamra@vinamra-mac ~ % htop
zsh: command not found: htop
vinamra@vinamra-mac ~ % brew install htop
==> Auto-updating Homebrew...
```

```
vinamra@vinamra-mac ~ % htop
```

Output->

```
0% 53.3% 4% 0.0%
1% 52.3% 5% 0.7%
2% 48.2% 4% 0.0%
3% 47.9% 7% 0.0%
Mem 15.45G/8.0G Tasks: 568, 107% thr, 0 kthr; 1 running
Swap 0K/0K Load average: 3.86 3.83 3.85
Uptime: 01:45:21

Main
PID USER PERM NI VIRT RES S CPU% MEM% TIME+ Command
153 vinamra 17 0 3910 21088 0 0.2 0.3 0:00.45 /usr/libexec/UserEventAgent (Aqua)
1811 vinamra 24 0 3930 9672 0 0.1 0.2 0:00.19 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
3692 vinamra 17 0 3920 28832 0 0.3 0.8 0:00.08 /usr/sbin/scrcapture -pdc -z keyboard.selection.clipboard
691 vinamra 17 0 3910 26400 0 0.4 0.1 0:01.51 /usr/libexec/sharingd
626 vinamra 24 0 3910 26760 0 0.3 0:01.48 /usr/libexec/rapportd
1388 vinamra 0 0 3930 92160 0 0.1 0:00.53 /System/Applications/Notes.app/Contents/MacOS/Notes
2498 vinamra 24 0 3920 4632 0 0.1 0:00.08 htop
1769 vinamra 17 0 3770 4764 0 0.5 0:01.87 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework/Versions/145.0.7632.160/Helpers/Google Chrome Helper (Renderer).app/Contents/MacOS/Google Chrome
372 vinamra 17 0 3920 29392 0 0.4 0:00.33 /System/Library/CoreServices/loginwindow.app/Contents/MacOS/loginwindow console
1808 vinamra 0 0 4300 98.38 0 0.2 0:01.18 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework/Versions/145.0.7632.160/Helpers/Google Chrome Helper.app/Contents/MacOS/Google Chrome Helper --t
619 vinamra 24 0 3910 26840 0 0.3 0:00.38 /usr/libexec/nsurlsessiond
3693 vinamra 17 0 3920 26512 0 0.3 0:00.08 /System/Library/CoreServices/scrcaptureui.app/Contents/MacOS/scrcaptureui
752 vinamra 17 0 3910 22160 0 0.3 0:00.23 /usr/libexec/networkserviceproxy
607 vinamra 17 0 3910 11280 0 0.1 0:00.19 /System/Library/PrivateFrameworks/AssetCacheServices.framework/Versions/A/XPCServices/AssetCacheLocatorService.xpc/Contents/MacOS/AssetCacheLocatorService --t
```

Step 2: Disk Usage Monitoring

```
vinamra@vinamra-mac ~ % df -h
Filesystem      Size  Used Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1  228Gi  12Gi   26Gi    32%   404k  278M    0%      /
devfs            197Ki  197Ki    0Bi   100%    680    0  100%     /dev
/dev/disk3s6     228Gi   24Ki   26Gi    1%    0  278M    0%     /System/Volumes/VM
/dev/disk3s2     228Gi  11Gi   26Gi   31%   1.5k  278M    0%     /System/Volumes/Preboot
/dev/disk3s4     228Gi  680Mi   26Gi    3%   311  278M    0%     /System/Volumes/Update
/dev/disk1s2     500Mi  6.0Mi  482Mi    2%    1  4.9M    0%     /System/Volumes/xarts
/dev/disk1s1     500Mi  6.2Mi  482Mi    2%    40  4.9M    0%     /System/Volumes/iSCPreboot
/dev/disk1s3     500Mi  724Ki  482Mi    1%    65  4.9M    0%     /System/Volumes/Hardware
/dev/disk3s5     228Gi  175Gi   26Gi   87%   5.0M  278M    2%     /System/Volumes/Data
map auto_home    0Bi    0Bi    0Bi  100%    0    0      -      /System/Volumes/Data/home
/dev/disk3s1     228Gi  12Gi   26Gi   32%   406k  278M    0%     /System/Volumes/Update/mnt1
```

Step 3: Check Folder Size (du)

```
vinamra@vinamra-mac ~ % du -s *
2216 Applications
776 CascadeProjects
```

Step 4: Monitor Running Processes

```
vinamra@vinamra-mac ~ % top
```

```
Processes: 565 total, 6 running, 2 stuck, 557 sleeping, 2292 threads
Load Avg: 3.81, 3.58, 3.09 CPU usage: 10.24% user, 15.03% sys, 73.70% idle SharedLibs: 302M resident, 74M data, 20M linkedin, MemRegions: 173368 total, 1635M resident, 282M private, 965M shared.
PhysMem: 645M used (1648M wired, 1550M compressed), 1953M unused, VM: 220T vsize, 4221M framework vsize, 0(0) swaptin, 0(0) swapouts, Network: packets: 78153/7861K in, 441216/125M out, Disk: 1578541/14G read, 414448/4896M written.

PID COMMAND %CPU TIME 0TH 1TH 2TH 3TH 4TH 5TH 6TH 7TH 8TH 9TH 10TH 11TH 12TH 13TH 14TH 15TH 16TH 17TH 18TH 19TH 20TH 21TH 22TH 23TH 24TH 25TH 26TH 27TH 28TH 29TH 30TH 31TH 32TH 33TH 34TH 35TH 36TH 37TH 38TH 39TH 40TH 41TH 42TH 43TH 44TH 45TH 46TH 47TH 48TH 49TH 50TH 51TH 52TH 53TH 54TH 55TH 56TH 57TH 58TH 59TH 60TH 61TH 62TH 63TH 64TH 65TH 66TH 67TH 68TH 69TH 70TH 71TH 72TH 73TH 74TH 75TH 76TH 77TH 78TH 79TH 80TH 81TH 82TH 83TH 84TH 85TH 86TH 87TH 88TH 89TH 90TH 91TH 92TH 93TH 94TH 95TH 96TH 97TH 98TH 99TH 100TH
153 vinamra 17 0 3910 21088 0 0.2 0.3 0:00.45 /usr/libexec/UserEventAgent (Aqua)
1811 vinamra 24 0 3930 9672 0 0.1 0.2 0:00.19 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
3692 vinamra 17 0 3920 28832 0 0.3 0.8 0:00.08 /usr/sbin/scrcapture -pdc -z keyboard.selection.clipboard
691 vinamra 17 0 3910 26400 0 0.4 0.1 0:01.51 /usr/libexec/sharingd
626 vinamra 24 0 3910 26760 0 0.3 0:01.48 /usr/libexec/rapportd
1388 vinamra 0 0 3930 92160 0 0.1 0:00.53 /System/Applications/Notes.app/Contents/MacOS/Notes
2498 vinamra 24 0 3920 4632 0 0.1 0:00.08 htop
1769 vinamra 17 0 3770 4764 0 0.5 0:01.87 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework/Versions/145.0.7632.160/Helpers/Google Chrome Helper (Renderer).app/Contents/MacOS/Google Chrome
372 vinamra 17 0 3920 29392 0 0.4 0:00.33 /System/Library/CoreServices/loginwindow.app/Contents/MacOS/loginwindow console
1808 vinamra 0 0 4300 98.38 0 0.2 0:01.18 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework/Versions/145.0.7632.160/Helpers/Google Chrome Helper.app/Contents/MacOS/Google Chrome Helper --t
619 vinamra 24 0 3910 26840 0 0.3 0:00.38 /usr/libexec/nsurlsessiond
3693 vinamra 17 0 3920 26512 0 0.3 0:00.08 /System/Library/CoreServices/scrcaptureui.app/Contents/MacOS/scrcaptureui
752 vinamra 17 0 3910 22160 0 0.3 0:00.23 /usr/libexec/networkserviceproxy
607 vinamra 17 0 3910 11280 0 0.1 0:00.19 /System/Library/PrivateFrameworks/AssetCacheServices.framework/Versions/A/XPCServices/AssetCacheLocatorService.xpc/Contents/MacOS/AssetCacheLocatorService --t
```

Step 5: Identify Resource-Intensive Processes

```
vinamra@vinamra-mac ~ % ps aux | sort -nrk 3 | head
root      527   0.0  0.1 418363792   9040  ??  Ss   8:06PM   55:49.31 /usr/libexec/InternetSharing
root      313  38.0  0.3 418444448   25456  ??  Rs   8:06PM   41:00.10 /usr/libexec/configd
root      378  35.4  0.3 418371808   26624  ??  Rs   8:06PM   34:54.14 /usr/libexec/airportd
root      442  19.6  0.1 418366560   10768  ??  Ss   8:06PM   11:11.85 /System/Library/Frameworks/Security.framework/Versions/A/XPCServices/authd.xpc/Contents/MacOS/authd_mdnresponder
_mdnresponder 457   7.2  0.1 418366256   11184  ??  Us   8:06PM   7:50.74 /usr/sbin/mdnsResponder
vinamra   583   6.6  0.2 418368080   20016  ??  U    8:06PM   7:11.92 /usr/libexec/UserEventAgent (Aqua)
_trustd   410   4.1  0.2 418374560   17552  ??  Ss   8:06PM   4:33.09 /usr/libexec/trustd
vinamra   667   3.8  0.4 418363200   35904  ??  S    8:06PM   0:00.55 /System/Library/PrivateFrameworks/CloudKitDaemon.framework/Support/cloudud
root      1      0.2  0.2 411895920   17088  ??  Ss   8:06PM   3:30.20 /sbin/launchd
vinamra   629   2.3  0.2 418373344   16768  ??  S    8:06PM   0:05.87 /usr/libexec/trustd --agent
```

Step 6: Save System Metrics to Log File

```
vinamra@vinamra-mac ~ % touch system_monitor.log
vinamra@vinamra-mac ~ % echo "System Report - $(date)" >> system_monitor.log
vinamra@vinamra-mac ~ % top -l 1 | head -n 10 >> system_monitor.log
vinamra@vinamra-mac ~ % df -h >> system_monitor.log
vinamra@vinamra-mac ~ % ps aux | sort -nrk 3 | head >> system_monitor.log
vinamra@vinamra-mac ~ % cat system_monitor.log
System Report - Sat Mar 14 22:05:35 IST 2026
Processes: 562 total, 3 running, 1 stuck, 658 sleeping, 2325 threads
2026/03/14 22:05:35
Load Avg: 3.21, 3.23, 3.43
CPU usage: 11.88% user, 25.17% sys, 62.93% idle
SharedLibs: 452M resident, 79M data, 22M linkedit.
MemRegions: 19384 total, 19404 resident, 214M private, 1280M shared.
PhysMem: 701M used (1538M wired, 1428M compressor), 582M unused.
VM: 2281 vsize, 4922M framework vsize, 0(0) swaptins, 0(0) swapouts.
Networks: packets: 994931/860M in, 456442/154M out.
Disks: 1580062/14G read, 447338/5224M written.
Filesystem      Size      Used      Avail Capacity  iused      ifree  Miused      Mifree  Mounted on
/dev/disk3s1s1  228Gi    12Gi    216Gi    5%      32k      484k  288M      8%      /
devfs           196Ki    196Ki    0Bi    100%     678      0    100%     0%      /dev
/dev/disk3s6    228Gi    24Ki    228Gi    1%       0      0      288M     0%      /System/Volumes/NM
/dev/disk3s2    228Gi    11Gi    217Gi    30%     1.5k    288M     0%      /System/Volumes/Preboot
/dev/disk3s4    228Gi    688Mi    227Gi    3%     311    288M     0%      /System/Volumes/Update
/dev/disk3s2    580Mi    6.9Mi    573Mi    1%       1      4.9M     0%      /System/Volumes/Asrts
/dev/disk1s1    580Mi    6.2Mi    574Mi    1%       40      4.9M     0%      /System/Volumes/ISCSPreboot
/dev/disk1s3    580Mi    724Ki    579Mi    1%       65      4.9M     0%      /System/Volumes/Hardware
/dev/disk3s5    228Gi    178Gi    110Gi    77%     5.8k    288M     2%      /System/Volumes/Data
map auto_home   0Bi     0Bi     0Bi    100%     0       0      0%      - /System/Volumes/Data/home
/dev/disk3s1    228Gi    12Gi    216Gi    5%     484k    288M     0%      /System/Volumes/Update/mnt1
root           527   0.0  0.1 418363792   9056  ??  Rs   8:06PM   59:31.99 /usr/libexec/InternetSharing
root           378  38.0  0.3 418372368   26880  ??  Rs   8:06PM   37:27.55 /usr/libexec/airportd
root           313  37.8  0.4 418449568   32112  ??  Us   8:06PM   43:41.62 /usr/libexec/configd
root           442   0.0  0.1 418366560   10760  ??  Ss   8:06PM   11:56.30 /System/Library/Frameworks/Security.framework/Versions/A/XPCServices/authd.xpc/Contents/MacOS/authd
_mdnresponder 457   6.6  0.1 418366256   11248  ??  Us   8:06PM   8:28.04 /usr/sbin/mdnsResponder
vinamra        583   6.6  0.2 418368080   20544  ??  R    8:06PM   7:39.95 /usr/libexec/UserEventAgent (Aqua)
_trustd        410   3.7  0.2 418374560   17456  ??  Ss   8:06PM   4:58.87 /usr/libexec/trustd
root           1      0.2  0.2 411895920   17216  ??  Ss   8:05PM   3:52.61 /sbin/launchd
vinamra       1448   2.5  3.7 1865731920  311392  ??  S    8:47PM   10:09.53 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/145.0.7632.160/Helpers/Google Chrome Helper (Renderer).app/Content
s/MacOS/Google Chrome Helper (Renderer) --type=renderer --metrics-client-id=9240807-b997-4511-b0a4-8573deed388e --origin-trial-disabled=features.CanvasTextNgWebAssemblyCustomDecorators --lang=en-US --no-raster-threads=4 --enable-peer-copy
--enable-gpu-memory-buffer-compositor-resources --enable-main-frame-before-activation --renderer-client-id=150 --time-ticks-at-unix-epoch=-1773498948722821 --launch-time-ticks=2589661432 --shared-files .metrics-shmem-handle=1752396122,r,150
840185328545805,17553088258566104016,2097152 --field-trial-handle=1718379636,r,8115985567589872685,13721795924531789912,262144 --variations-seed-version=20260313-170042.569000-production --trace-process-track-uid=3190789126868148844 --seatb
all-client=232
networkd       462   1.7  0.3 418366800   24272  ??  Ss   8:06PM   2:21.35 /usr/libexec/symptomsd
```

Challenges Encountered

1. macOS does not support **apt package manager**, so **Homebrew** was used instead.
2. Some Linux commands behave slightly differently on macOS.

Task 2: User Management and Access Control

Step 1: Create Workspace Directories

```
vinamra@vinamra-mac ~ % mkdir -p ~/Sarah/workspace
vinamra@vinamra-mac ~ % mkdir -p ~/mike/workspace
vinamra@vinamra-mac ~ % ls -ld ~/Sarah/workspace
drwxr-xr-x  2 vinamra  staff   64 Mar 15 19:32 /Users/vinamra/Sarah/workspace
vinamra@vinamra-mac ~ % ls -ld ~/mike/workspace
drwxr-xr-x  2 vinamra  staff   64 Mar 15 19:32 /Users/vinamra/mike/workspace
```

Step 2: Set Secure Permissions

```
vinamra@vinamra-mac ~ % chmod 700 ~/Sarah/workspace
vinamra@vinamra-mac ~ % chmod 700 ~/mike/workspace
vinamra@vinamra-mac ~ % ls -ld ~/Sarah/workspace
drwx-----  2 vinamra  staff   64 Mar 15 19:32 /Users/vinamra/Sarah/workspace
vinamra@vinamra-mac ~ % ls -ld ~/mike/workspace
drwx-----  2 vinamra  staff   64 Mar 15 19:32 /Users/vinamra/mike/workspace
```

Step 3: Simulate User Accounts

```
vinamra@vinamra-mac ~ % mkdir ~/users
vinamra@vinamra-mac ~ % mkdir ~/users/Sarah
vinamra@vinamra-mac ~ % mkdir ~/users/mike
vinamra@vinamra-mac ~ % ls ~/users
Sarah  mike
```

Step 4: Password Policy

macOS password policies are managed via **System Settings**

Example policy documentation:

1. Passwords must contain at least 8 characters.
2. Must include uppercase and lowercase letters.
3. Must include at least one numeric digit.
4. Passwords expire every 30 days.
5. Users cannot reuse their last three passwords.

Step 5: Access Control Verification

```
vinamra@vinamra-mac ~ % cd ~/Sarah/workspace
vinamra@vinamra-mac workspace % touch testfile.txt
vinamra@vinamra-mac workspace % ls ~/Sarah/workspace
testfile.txt
```

Step 6: Disk Ownership Check

```
vinamra@vinamra-mac workspace % whoami
vinamra
```

Task 3: Backup Configuration for Web Servers

Step 1: Create Server Directories

```
vinamra@vinamra-mac workspace % mkdir -p ~/servers/apache/httpd
vinamra@vinamra-mac workspace % mkdir -p ~/servers/apache/html
vinamra@vinamra-mac workspace % mkdir -p ~/servers/nginx/nginx
vinamra@vinamra-mac workspace % mkdir -p ~/servers/nginx/html
vinamra@vinamra-mac workspace % ls ~/servers
apache  nginx
```

Step 2: Create Backup Folder

```
vinamra@vinamra-mac workspace % mkdir ~/backups
```

Step 3: Create Apache Backup Script (Sarah)

```
[vinamra@vinamra-mac workspace % nano apache_backup.sh
[vinamra@vinamra-mac workspace % cat apache_backup.sh
#!/bin/bash

DATE=$(date +%F)

BACKUP_FILE="/backups/apache_backup_$DATE.tar.gz"

tar -czf $BACKUP_FILE /etc/httpd /var/www/html

echo "Backup created: $BACKUP_FILE"

tar -tzf $BACKUP_FILE > /backups/apache_backup_log.txt
[vinamra@vinamra-mac workspace % ./apache_backup.sh
zsh: permission denied: ./apache_backup.sh
[vinamra@vinamra-mac workspace % chmod +x apache_backup.sh
[vinamra@vinamra-mac workspace % ls -l apache_backup.sh
-rwxr-xr-x  1 vinamra  staff  222 Mar 15 20:25 apache_backup.sh
```

Step 4: Create Nginx Backup Script (Mike)

```
[vinamra@vinamra-mac workspace % nano nginx_backup.sh
[vinamra@vinamra-mac workspace % cat nginx_backup.sh
#!/bin/bash

DATE=$(date +%F)

BACKUP_FILE="/backups/nginx_backup_$DATE.tar.gz"

tar -czf $BACKUP_FILE /etc/nginx /usr/share/nginx/html

echo "Backup created: $BACKUP_FILE"

tar -tzf $BACKUP_FILE > /backups/nginx_backup_log.txt
[vinamra@vinamra-mac workspace % chmod +x nginx_backup.sh
[vinamra@vinamra-mac workspace % ls -l nginx_backup.sh
-rwxr-xr-x  1 vinamra  staff  228 Mar 15 20:37 nginx_backup.sh
```

Step 5: Run Backup Scripts

Script failed

```
[vinamra@vinamra-mac workspace % ./apache_backup.sh
tar: Failed to open '/backups/apache_backup_2026-03-15.tar.gz'
Backup created: /backups/apache_backup_2026-03-15.tar.gz
./apache_backup.sh: line 11: /backups/apache_backup_log.txt: No such file or directory
```

Fixed script

```
vinamra@vinamra-mac workspace % nano apache_backup.sh
vinamra@vinamra-mac workspace % cat apache_backup.sh
#!/bin/bash

DATE=$(date +%F)

BACKUP_FILE=~/backups/apache_backup_$DATE.tar.gz

tar -czf $BACKUP_FILE ~/servers/apache

echo "Backup created: $BACKUP_FILE"

tar -tzf $BACKUP_FILE > ~/backups/apache_backup_log.txt
```

```
[vinamra@vinamra-mac workspace % nano nginx_backup.sh
[vinamra@vinamra-mac workspace % cat nginx_backup.sh
#!/bin/bash

DATE=$(date +%F)

BACKUP_FILE=~/backups/nginx_backup_$DATE.tar.gz

tar -czf $BACKUP_FILE ~/servers/nginx

echo "Backup created: $BACKUP_FILE"

tar -tzf $BACKUP_FILE > ~/backups/nginx_backup_log.txt
```

Script worked, backup created

```
[vinamra@vinamra-mac workspace % ./apache_backup.sh
tar: Removing leading '/' from member names
Backup created: /Users/vinamra/backups/apache_backup_2026-03-15.tar.gz
```

```
[vinamra@vinamra-mac workspace % ./nginx_backup.sh
tar: Removing leading '/' from member names
Backup created: /Users/vinamra/backups/nginx_backup_2026-03-15.tar.gz
```

Step 6: Verify Backup File

```
vinamra@vinamra-mac workspace % ls ~/backups
apache_backup_2026-03-15.tar.gz  apache_backup_log.txt          nginx_backup_2026-03-15.tar.gz  nginx_backup_log.txt
```

Step 7: Verify Backup Integrity

```
[vinamra@vinamra-mac workspace % tar -tzf ~/backups/apache_backup_2026-03-15.tar.gz
Users/vinamra/servers/apache/
Users/vinamra/servers/apache/html/
Users/vinamra/servers/apache/httpd/
```

Step 7: Add Cron Job

```
[vinamra@vinamra-mac workspace % crontab -e
crontab: no crontab for vinamra - using an empty one
crontab: installing new crontab
[vinamra@vinamra-mac workspace % crontab -l
0 0 * * 2 /Users/vinamra/apache_backup.sh
0 0 * * 2 /Users/vinamra/nginx_backup.sh
```

Challenges Encountered

1. While creating directories like `/etc/httpd` and `/etc/nginx`, the system returned **permission denied errors** because these directories belong to the system root. So tried simply created server folders inside your home directory
2. Backup Script Failed Due to Missing Backup Directory
3. Backup script save & exit in nano Editor
4. Cron Job Editing with Vim Editor